🖱️ Air Mouse Concept

An Air Mouse lets you control a cursor by moving your hand in the air, using motion sensors like an accelerometer or gyroscope instead of a traditional mouse.

🔧 COMPONENTS NEEDED

Arduino Uno / Nano / ESP32 (microcontroller)
MPU6050 (accelerometer + gyroscope)
Bluetooth module (HC-05/HC-06) or USB cable for PC connection
Battery (Li-ion or Li-Po)
Breadboard & jumper wires
Optional: 3D printed case for handheld comfort

⚙ WORKING PRINCIPLE

Motion Capture – MPU6050 detects tilt, rotation, and acceleration.
Data Processing – Arduino converts sensor readings into cursor movement values.
Communication – Data sent to PC via Bluetooth or USB serial.
Cursor Control – A small program on PC (e.g., Processing, Python, or custom driver) maps these values to mouse movements and clicks.

💻 Example Arduino Code (MPU6050 + Serial)
Here’s a minimal sketch to read sensor data:

Cpp#include <Wire.h>
#include <MPU6050.h>

MPU6050 mpu;

void setup() {
  Serial.begin(9600);
  Wire.begin();
  mpu.initialize();
  if (!mpu.testConnection()) {
    Serial.println("MPU6050 connection failed!");
  }
}

void loop() {
  int16_t ax, ay, az, gx, gy, gz;
  mpu.getMotion6(&ax, &ay, &az, &gx, &gy, &gz);

  Serial.print("Accel: ");
  Serial.print(ax); Serial.print(", ");
  Serial.print(ay); Serial.print(", ");
  Serial.println(az);

  delay(50);
}

🛠️ Next Steps

Map ax and ay values to cursor X/Y movement.
Add gesture detection for clicks (e.g., quick shake for left click).
Use Processing or Python PyAutoGUI to move the cursor based on incoming serial data.

If you want, I can prepare a full Arduino + Processing code so your Air Mouse works instantly with your PC — including click gestures and smooth cursor movement.
Do you want me to make that?